

AI Automaton: AI Systems Intended to Imitate Humans

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There is a proliferation of AI systems designed to mimic people's behavior, work, abilities, likenesses, or humanness—systems we dub *AI automaton*s. Individuals, groups, or generic humans are simulated to produce creative work in their styles, respond to surveys in their places, probe how they would use a new system before deployment, provide users with assistance and companionship, and anticipate their possible future behavior and interactions with others, just to name a few applications. However, the research, design, deployment, and availability of such AI systems have prompted growing concerns about a wide range of possible legal, psychological, social, and other types of harms. In this paper, we seek 1) to facilitate productive discussions about *whether*, *when*, and *how* to design and deploy such systems, and 2) to help chart the current landscape of existing and prospective *AI automaton*s. To do so, we tease apart determinant design axes and considerations to aid reflections and deliberations about whether and how design choices along these axes could *mitigate*—or instead *exacerbate*—harms that the development and use of *AI automaton*s might give rise to. Through a synthesis of related literature and extensive examples of existing AI systems intended to mimic humans, we developed a conceptual framework that foregrounds key axes of design variations and provides analytical scaffolding to foster greater recognition of a) the design choices available to developers and researchers, as well as of b) the possible ethical implications these design choices might have.

CCS Concepts: • **Human-centered computing** → **HCI theory, concepts and models**; **Interactive systems and tools**; • **Social and professional topics**; • **Computing methodologies** → *Modeling and simulation*; *Artificial intelligence*;

Additional Key Words and Phrases: anthropomorphism, anthropomorphic AI, AI automaton, impacts

ACM Reference Format:

Alexandra Olteanu, Solon Barocas, Su Lin Blodgett, Lisa Egede, Alicia DeVrio, and Myra Cheng. 2026. *AI Automaton: AI Systems Intended to Imitate Humans*. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*, June 25–28, 2026, Montreal, QC, Canada. ACM, New York, NY, USA, 39 pages. <https://doi.org/10.1145/3805689.3812377>

1 Introduction

There is a fast-growing number of examples where AI systems are developed or used to imitate humans. These include cases where the likenesses and voices of deceased or missing children were reenacted to help narrate their stories of abuse and violence [122], such as that of a “17-month-old [who] died in 2007 following months of physical abuse” [106], or cases of AI characters meant to depict made-up intersectional identities, such as “Liv’

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FAccT '26, Montreal, QC, Canada

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ACM ISBN 979-8-4007-2596-8/2026/06

<https://doi.org/10.1145/3805689.3812377>

portraying a ‘proud Black queer momma of 2 & truth-teller’” [187]. There is also “[a]n array of popular apps [that] are offering AI companions [...] who are spinning up AI girlfriends, AI husbands, AI therapists—even AI parents” [252]. Some developers seek to mimic specific individuals [e.g., 33, 154, 156, 174, 230], while others only aim to imbue systems with more general human-like characteristics [e.g., 140, 244].

Indeed, growing beliefs that AI systems will be or are already human-like and able to replicate a wide range of human abilities or likenesses [e.g., 47, 56, 82, 94, 172]—AI systems we dub *AI automatons*¹ to emphasize their *mechanical nature*—have led both to growing interest in developing such systems [e.g., 112, 188, 235], as well as to growing concerns about their potential to replace humans in various jobs, about the potential emotional toll from interacting with seemingly human-like systems, or about adverse impacts to humans in other more or less anticipated ways [e.g., 7, 30, 33, 38, 42, 57, 83, 154, 185, 252]. This trend is far from new, with the field of AI itself guided by the question of whether computational systems are capable of thought or at least of faithfully imitating humans—also known as Turing’s *imitation game* [98]—an aim which has a long history in fields like gaming [e.g., 12, 152, 167], human-robot interaction [e.g., 60, 79], and animation and motion pictures [e.g., 84, 177].

What is perhaps *new* is the increasing feasibility and availability of AI systems that could be used to simulate specific individuals or highly realistic human-like entities that are able to engage in increasingly autonomous and open-ended interactions with others, coupled with a growing ubiquity of such systems across a wider and wider range of applications, both of which are propelled by expanding and increasingly pervasive claims about, and perceptions of, AI systems’ capabilities. The growing possibility of developing highly accurate simulations of individuals, in particular, has led to examinations of their ethical and social implications [e.g., 33, 122, 154, 162, 174], and of the possibility of these simulations being used to replace humans [e.g., 7, 261, 275]. While critically important, these early efforts tend to focus on either a specific class of AI automatons or a specific class of concerns, and do not *examine what and how various design choices may amplify different types of concerns and risks—for which we seek to provide an analytical foundation in this paper*.

Contributions. In this work, we develop a conceptual framework that maps key design considerations when building and deploying AI automatons (§3), highlighting possible adverse impacts different design choices might have. In doing so, we focus on the *intended* goals for such AI automatons, rather than attempting to speculate about what AI automatons can or cannot do or about the incentives and motivations of those developing these systems (beyond the stated goals for what the AI automatons are intended for). In other words, we focus on what AI automatons are developed, deployed, used, or intended to be used for. Our aims are two-fold: 1) chart the current landscape of existing and prospective *AI automatons*, and 2) provide analytical scaffolding and a foundation for a) discussions about *whether*, *when*, and *how* to design and deploy such systems, and for b) future examinations of the adverse impacts certain types of configurations might have. Our framework does so in a few ways: First, a greater recognition of possible design choices can help developers and researchers² of AI automatons both to be more intentional and explicit about the choices they make for *how* to design and *when* to deploy AI automatons, as well as to discern potential adverse impacts that different choices and their interactions may have. Second, added clarity about different choices’ impacts can encourage them to consider alternative choices—for *how* to design and deploy AI automatons—that may help mitigate these impacts, or identify redlines to guide decisions about *whether* to build certain types of AI automatons. Finally, by providing a common, systematized terminology, our framework can facilitate dialogue among developers and researchers, enabling more transparent, standardized documentation and reporting of AI automatons’ characteristics.

¹An *AI automaton* is a AI system that is “relatively self-operating” or that is “designed to follow automatically a predetermined sequence of operations or respond to encoded instructions” [179] in order to reproduce or mimic humans or their characteristics and behavior. As a result, any AI system intended to *simulate* humans by reproducing their characteristics and behaviors constitutes an AI automaton.

²We use *developers and researchers* as a shorthand for stakeholders that make decisions about the design, development, or deployment of AI automatons. As we will see in §2.3, in our framework these stakeholders also take the role of *operators* or *interactors*.

2 Background & Related Work

AI systems are increasingly anthropomorphic [11, 47, 149, 171, 225]—described or perceived as human-like. Anthropomorphism can be *by design*, often by incorporating human-like features into systems, e.g., avatars with different skin colors or hairstyles [105], or robots with facial features [239] or producing human sounds and gestures [161]. Such design choices may be motivated by desires to increase users’ engagement, comfort, familiarity, or trust [130, 139, 215, 219, 254], improve user experiences [109, 274, 277], or encourage consumer engagement [196] and consumption [101, 103, 197], though anthropomorphization can also backfire [178]. AI systems may also be anthropomorphic even when *not intentionally designed for*. For example, language use, until recently solely a human activity and made possible for AI systems by training on large quantities of human-produced language, can readily give rise to perceptions of human-likeness [69].

2.1 Simulating Humans

To appear human-like, however, AI systems need to reproduce or appear to reproduce human characteristics.

Simulating individuals. For instance, simulations have been developed to target a wide array of individuals, including models personalized to specific chess players to predict their next moves [175], simulating individual Supreme Court justices to predict future decisions [102], and simulating users for sending emails, “match[ing] the voice and tone in the emails you’ve already sent, applying that to everything [the model] creates” [257]. Particularly when they include a visual component, such simulations are often also referred to as *deepfakes* [33, 191], with a 2024 CBS news report highlighting that there were “more than 21,000 deepfake pornographic videos online—up more than 460% over the year prior” [17]. Simulations may target people no longer alive, including loved ones as well as public figures [123, 185]. Emerging applications also include those aimed at simulating many individuals at once, such as for pilot studies [222] and polls [288]. Park et al. [201] simulate “attitudes and behaviors of 1,052 real individuals.” Other simulations may target fictional individuals—e.g., models role-playing as specific characters [263]—as well as entirely new characters, “offering AI companions to millions of [...] users” [252].

Simulating groups. Simulations may also target members of social groups, ranging from social or professional roles to demographic groups. For example, Qian et al. [212] develop agents in roles like programmers and test engineers for software development, and Sun et al. [247] develop a legal consultation system with agents in roles like receptionists and lawyers. Argyle et al. [15] construct prompts with demographic information to see if a model’s output reflects response distributions (for e.g., surveys) “each aligned with a real human sub-population,” while Lee et al. [155] investigate whether LLMs conditioned on demographics can simulate responses to climate change surveys. Basoah et al. [22] examine user perceptions of systems using features of two English sociolects, which some participants perceived as more human-like than a system using Standard American English. Other simulations of members of demographic groups include AI social media accounts, like “Grandpa Brian,” a Meta account which “described itself ... as an African-American retired entrepreneur” and whose bio is an “entirely fictionalized biography based on a composite of real African American elders’ lives” [187].

Simulating human phenomena & interactions. An increasing number of applications also seek to simulate people in order to study human phenomena, including people’s beliefs and attitudes [201]; social dynamics and interactions in simulated communities and social networks [86, 214, 253, 262, 281]; and human decision-making across a range of settings, such as resource allocation [132] and government responses to public disaster [276]. See Mou et al. [188] for a survey.

2.2 Growing Concerns

As anthropomorphic AI systems have proliferated, work has also emerged raising critical concerns about them. Scholars have long problematized “relational artifacts [...] *specifically designed to make people feel understood*,

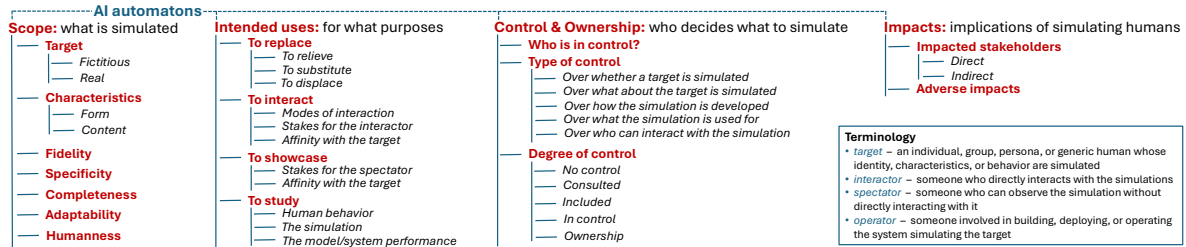


Fig. 1. Overview of the design considerations covered by our conceptual framework. For a more detailed breakdown of the considerations we identified for AI automatons (along with examples), see Table 1 in Appendix A.

[artifacts that] are still without understanding” [255, emphasis original], as they lack the human understanding required for the relationships people try to create with them. Moreover, encouraging users to relate to AI systems as if they are human can cause over-reliance on system outputs [2, 71]—leading to exaggerated perceptions of AI capabilities as well as distorting moral judgments about responsibility [207, 229]—and negatively impact critical thinking abilities [88]. Such systems can also prevent “users from assuming certain roles themselves, or [...] from questioning the need for certain roles in the first place” [166]. In the long term, such systems may lead to long-term emotional dependence [85], and their capacity to express feelings they cannot have may devalue expressions of genuine emotion and erode our ability to bond with each other [208]. Their perceived trustworthiness [14]—potentially heightened by misuse of users’ personal information [166]—may enable increased user deception, manipulation, and exploitation [266], and combined with their increased ubiquity risks gradual acclimatization that may facilitate increased public acceptance of potentially unethical uses of AI [215].

As they are designed to imitate humans, AI automatons are likely to be perceived as human-like and thus give rise to the same concerns as anthropomorphic systems more broadly. However, as systems *explicitly* designed to simulate people’s behavior, work, abilities, or likenesses, they can give rise to additional concerns, particularly about harms to those whose characteristics they purport to reproduce. Some of these concerns involve practical challenges; for example, Agnew et al. [7] identify limitations of current such systems, including their tendency to make mistakes and to reproduce dominant perspectives rather than those of the people they are intended to replace. Wang et al. [261] argue that LLMs’ tendencies to *misportray* demographic groups (generate out-group rather than in-group members’ perspectives) and *flatten* groups (treat groups as monoliths, erasing heterogeneity and neglecting intersectional identities) make them unsuitable as replacements for human participants.

Beyond these concerns—which may be overcome with modeling advancements—these and other works have also identified *concerns fundamental to the act of simulating people*. Agnew et al. and Wang et al. note how replacing study participants reproduces minoritized groups’ exclusion from decision-making and moves away from the meaningful sharing of power that is core to visions of inclusion. Wang et al. remark that such simulations also risk essentializing identity by treating identities as “rigid and innate.” McIlroy-Young et al. [174] characterize normative concerns arising from *mimetic models*—generative and interactive models simulating specific people. Lee et al. [154] investigate users’ and targets’ perceptions of simulations (*AI clones*), identifying concerns ranging from misrepresentation to replacement and exploitation.

2.3 Concepts & Working Terminology

Drawing on McIlroy-Young et al. and others using similar terminologies [122, 154, 156], to differentiate between different types of stakeholders our framework considers the following stakeholder roles:³ (1) *target* – an individual,

³Unlike [174], we do not distinguish between the builders of the systems and the operators of the systems, and by definition, spectators are different from operators and interactors. We also use a more expansive definition for the target, which does not need to be a specific individual.

group, persona, or generic human whose identity, characteristics, or behavior are simulated; (2) *interactor* – someone who directly interacts with the simulation; (3) *spectator* – someone who can observe the simulation without directly interacting with it; (4) *operator* – someone involved in the building, deployment, or operation of the system simulating the target. However, the stakeholder roles can overlap, with the same entity being able to take on multiple roles; for instance, the *interactor* can be the same as the *target*, such as when users interact with their own replicas [e.g., 154]. The same role can also be inhabited by multiple stakeholders; for instance, different aspects of the operator role might be under the purview of different organizational stakeholders such as developers, project managers, and executives [181].

3 Conceptual Framework for AI Automaton

Our aim is to identify key design considerations for AI automaton that might introduce new risks or heighten existing ones. To tease apart design considerations that influence perceptions of and interactions with AI automaton, and help determine the types of harms these systems may give rise to, we consider various aspects related to what a simulation is intended for, how these intended goals are accomplished, and who is influencing these decisions.

Methodological approach. Our conceptual framework grew out of a review of a purposive sample of related work on simulating humans [e.g., 7, 112, 154, 174], from which we identified an initial set of design considerations (*Step 1*).⁴ Specifically, to assemble this sample, we employed criterion-based purposive sampling [199, 204], and included only papers that matched the following *selection criteria*: they were concerned with 1) AI systems simulating humans or their characteristics, and/or 2) the risks and harms that these systems might give rise to. We then clustered the sets of design considerations mentioned in these papers in a bottom-up fashion by function (*Step 2*), arriving at three initial top-level categories (target, intended uses, impacts) and several subcategories (e.g., fidelity, replacement, interaction, stakeholders). To expand and refine these categories, we then followed an iterative, inductive-abductive approach [e.g., 121, 129, 169] that mixed considerations about *what-is*—how existing AI automaton are currently designed—and *what-might-be*—speculating about alternative ways AI automaton could be designed. To do so, we read broadly to identify (inductively, *Step 3*) recurring examples of AI automaton and related design considerations, and reflected (abductively, *Step 4*) on additional possibilities for AI automaton implied by existing work but perhaps not yet commonplace.

By moving from empirical observations—based on the examples we identified in the already reviewed literature—to hypothesizing about possible axes of variation, this approach enabled us to derive a more comprehensive, robust conceptual framework. Specifically, we iterated on and expanded the framework through collaborative discussion sessions with subsets of our research team where we considered additional literature identified in a snowball fashion—i.e., literature that was cited in (backward snowball sampling) or citing (forward snowball sampling) literature we already included, or literature that we found to be examining specific design considerations that we identified during the previous iterations, such as control or interaction (opportunity sampling [204]). We focused only on papers that matched the same *selection criteria* we used for the initial purposive sample. During each session, we supplemented the literature with examples of both existing—opportunistically identified during the study—and speculative applications, which we then used to probe whether the framework was missing key design variations and considerations. When we identified additional considerations, we expanded the framework and, if the same consideration emerged across multiple framework branches, we reorganized the framework to elevate the consideration as a separate dimension at the same level as the common root the branches shared (e.g., after several iterations control emerged as an important cross-cutting consideration for multiple use-scenarios).

Our resulting framework is organized along 4 design axes (Fig. 1): 1) what is simulated (*simulation scope*, §3.1), 2) for what purposes (*intended uses*, §3.2), 3) who controls what is simulated (*ownership & control*, §3.3), and 4) how

⁴Purposive sampling is a non-probability-based sampling method relying on researchers' expertise and judgment to purposefully identify and select information-rich cases (here papers) that appropriately support the study's goals [204].



Fig. 2. Simulations can range from targeting characteristics of specific individuals to more general group and even more general human characteristics (x-axis). They can also range from simulating single or multiple entities (y-axis).

the simulation impacts stakeholders (*adverse impacts*, §3.4). See Appendix B for a more structured, step-by-step overview of our process to identify design axes.

3.1 Scope: What Is Simulated and How?

Aspects related to *who* and *what* about them is simulated (§3.1.1), as well as *how* they are simulated (§3.1.2), govern perceptions of and concerns about systems simulating people’s work, abilities, behavior, likenesses, or humanness [e.g., 75, 154, 283]. For instance, *what* is simulated and *how* the simulation is accomplished can influence perceptions of uncanniness and discomfort [e.g. 80, 97, 154, 173], particularly when aspects unique to an individual are simulated, when the simulation outputs appear eerily similar to what a human might do or look like, or when they capture one’s characteristics with high fidelity. Such properties of a simulation may also exacerbate other concerns like concerns about privacy violations and lack of appropriate consent [30, 141, 154], or may threaten someone’s sense of identity and agency [7, 81, 154].

3.1.1 What is being simulated? This involves decisions about both *who* the target of the simulation is and *what about them* is simulated. This distinction can help clarify aspects related to who might be impacted by the simulation, and in what ways they might be impacted based on what about them is being mimicked by the AI automaton.

Target: who is simulated? Irrespective of the deployment context or scenario, the most foundational characteristics of an AI automaton—particularly as it pertains to determining if a simulation should even be developed, what the goals of the simulation should be, and how the simulation should be developed—are whether a person (*fictitious* or not) is simulated, and which of their characteristics or behaviors are simulated. AI automatons can range from simulating specific, *real* individuals [e.g., 20, 148, 154, 174]; to cases where the simulations target real individuals but also aim to imbue the automaton with skills or characteristics these individuals do not have [e.g., 154], or target realistic, yet fictitious individuals [e.g., 28, 55, 111, 187]; to cases where the simulation targets a group by simulating a target deemed prototypical for the group [e.g., 264, 287] or by simulating a population of targets that all belong to one or multiple groups [e.g., 201]; to cases where the aim is just to simulate something that appears to be human in some way [e.g., 94, 189, 232].

To operationalize such distinctions in what the target could be, we differentiate between i) simulations where the target is a specific individual (e.g., simulating someone’s writing style) from ii) simulations where the target is a group or a persona representative of the group (e.g., simulating how the average woman would answer a survey question), or one embodying characteristics specific to a group without necessarily being representative (e.g., simulating someone’s voice for certain attributes, such as “a female voice with a North American accent” [122]), and from iii) simulations targeting more generic human characteristics that are not considered specific to any individual or group (e.g., avatars expressing human emotions [34]).

These distinctions also map to a design spectrum (illustrated in Fig. 2) where a target can vary from specific individuals to general group characteristics to even more general human characteristics, and from representing a single entity to simulating multiple entities (e.g., a population of targets from the same group [e.g., 155]). These distinctions matter as concerns related to, e.g., deception, impersonation, or lack of consent and control over one's identity and likeness [e.g. 17, 74, 77, 137, 162, 272] may be exacerbated when simulating the decisions or behaviors of specific, recognizable people—such as those of a specific AI researcher like Geoffrey Hinton—versus when simulating a generic AI researcher, or only when embodying nonspecific human-like traits.

Another critical aspect in specifying a target is whether the target or some (or all) of their characteristics are i) fictitious (e.g., chatbots simulating human-like characters from popular games, films, or anime [252]) versus when they are ii) real (e.g., chatbots simulating real school shooters and their victims [74] or deceased loved ones [20, 185]), since such differences can also have drastically different ethical and social implications. Simulating fictitious characters that cannot reasonably be matched to a real person or group is less likely to raise concerns about identity fragmentation—i.e., when replicas of an individual threaten their perceived individuality—and objectification [154], about impersonation [26], or about privacy and lack of consent [125]. On the other hand, imbuing simulations of real people with characteristics those people do not have, or perfecting the characteristics they do have [e.g., 154, 233]—i.e., by setting the target as an altered, fictitious version of a real person—might increase concerns, such as about the displacement of those being reproduced [154]. Such simulations could also bring about relational or emotional harms [43, 65, 285] if users become emotionally dependent on the AI automaton or experience social withdrawal. Favoring simulations of fictitious rather than real people may not, however, be a way to completely sidestep the ethical issues raised by automats [e.g., 258].

Characteristics: what about the target is simulated? Furthermore, an AI automaton may be developed only to capture some of a target's characteristics, and may also aim to do so only for specific actions or tasks a target may undertake. An example could be an image generator developed to mimic one's drawing style [146], but which does so only for specific elements in generated images but not for the rest of the image (e.g., for flowers present in an image), or a model developed to simulate only how one would respond to a given set of questions or only one's visual likeness. Choices about different elements related to *which* of a target's characteristics are simulated and *how* they are simulated (§3.1.2) can be more or less likely to lead to a simulation being deemed e.g., uncanny or unsettling [80, 173], and are likely to govern perceptions about and interactions with the overall system in different ways [12]. For instance, simulating someone's voice may heighten impersonation or deception concerns [26, 80, 122, 272] more than how only simulating their written responses to a predefined set of questions would [e.g., 15].

Moreover, simulating both what someone might say in response to a question and how they would say it (including word choices, sentence structure, or voice and accent)—i.e., combining different types of characteristics the target might have—is more likely to exacerbate concerns about the system impersonating or even mocking the target [e.g., 10, 89, 108]. Since simulating *what* a target does or might do (e.g., what they would say) as opposed to simulating *how* the target appears or does something (e.g., how they might say it) may lead to different concerns, we also distinguish between aiming to simulate characteristics that are related to: i) form: when mimicking the likeness, appearance, or style of a target, versus ii) content: when mimicking what a target might say or do.

The choice of which characteristics to simulate also influences and is influenced by considerations related to how *coherent*, *realistic*, *naturalistic*, or *plausible* the simulation needs or is intended to be [e.g., 4, 72, 108, 153, 154, 168, 174]. For instance, the goal of simulating how a specific individual would respond to a question may require the answer be something the target might *plausibly* say, and that the way the response is formulated also *coherently* reflects their communication style [108, 154]. Another key determinant factor constraining which of the target's characteristics could even be simulated by a system is the *modality* of a system, or “the domain [the system] operates in and the types of behaviors it is designed to reflect” [174], which in turn can impact perceptions [150]. A system producing open-ended texts will be able to capture a different set of a target's characteristics than

one that outputs videos or one that only outputs answers to categorical questions. All these considerations also depend on design choices related to *how* to simulate the target and their characteristics, which we discuss next.

3.1.2 How are the target and their characteristics being simulated? Once settled on what should be reproduced about a target, there are also many design choices about the nature of the simulation itself, such as those related to the fidelity with which to reproduce the target's characteristics or to whether the simulation should remain faithful to a static snapshot of a target or whether it can evolve. The interplay between such design choices—and their interdependency with design choices about what to simulate (§3.1.1)—can further exacerbate concerns.

Fidelity: how well is the simulation intended to capture the target's characteristics? The fidelity or accuracy with which an AI automaton reproduces a target's characteristics is likely to impact the value or usefulness of these systems [e.g., 12, 15, 131, 174, 258]—that is, the more faithfully a system mimics a target, the better—as it directly relates to whether the simulation is or appears to be *coherent*, *realistic*, and *plausible*. While low-fidelity simulations can give rise to concerns related to misrepresentation, deception, or reputational harms when the simulation drifts away from appropriately and accurately representing the target [28, 154, 174], other ethical or legal concerns are more likely to arise due to high-fidelity simulations [e.g., 40, 154]—particularly when the target has little control (§3.3) over whether and how their work, abilities, or likeness are simulated [e.g., 13, 133, 137, 154, 162]. For instance, high-fidelity simulations of an artist's style, work, or likeness are more likely to lead to copyright violations or infringement on their rights of publicity than low-fidelity simulations (e.g., simulating a generic female-sounding voice vs. Scarlett Johansson's voice [137]).

Specificity: are the simulated characteristics unique to the target? In addition to how faithfully a target or their characteristics are simulated, how *unique* these characteristics are to a target—e.g., in a way that uniquely represents or identifies them, or reproduces unique or rare abilities—is also critical to consider, as the reproduction of such characteristics raises questions about one's ability to maintain their individuality [154], reputation (e.g., when a public figure's voice is simulated to spread defamatory content [122]), ability to capitalize on their own skills or talents (e.g., when reproducing an writer or artist's signature style [13, 146]), or ability to maintain control over their own name or image [e.g., 13, 23]. The mimicking of one's unique characteristics can also exacerbate privacy or impersonation risks [e.g., 18, 23, 216, 267], and both individuals and professional or cultural groups risk seeing their work devalued or losing part of their social capital or even livelihood [13, 146, 174].

Completeness: is the simulation intended to fully capture the target? How many of a target's characteristics are simulated, or whether the target is intended to be *simulated in its entirety*, is another consideration that determines not only simulations' deployment settings, but also how they are perceived and interacted with [154] and how *versatile* the resulting automaton is—e.g., the diversity of actions it can take [183]. Simulations intended to be highly detailed and elaborate, be exhaustive, or have high *generality*—capturing a substantial “breadth of scenarios and domains” [174]—will lead not only to more but also to heightened concerns, particularly when a system's reach is more extensive. For instance, systems designed to “visually resemble you, sound like you, and possess the knowledge you would want to carry into [a] meeting” [156]—requiring simulation of many characteristics—may yield different concerns than those simulating yes/no responses to survey questions [e.g. 155]. Highly complex simulations of individuals are more likely to trigger concerns about objectification, de-humanization, displacement, or loss of individuality [154], and such concerns might be further exacerbated depending on the fidelity of those simulations [107, 154, 174].

Adaptability: is the simulation intended to evolve or adapt? While for some settings AI automatons may be intended to remain *static* or reflect fixed snapshots of a target (e.g., cloning one's younger self to talk to them [260]), other settings may require automatons to *evolve* 1) based on interactions, feedback, or new information (e.g., by learning from interactions [192, 265]), or 2) according to the target's own evolving self (e.g., to maintain accurate representations of the target [154]). But a static snapshot or one that evolves separately from

the target may misrepresent the target by presenting stale or inauthentic versions of them [154, 164]. Adaptation could also involve considerations about whether the automaton can adapt based on context (e.g., changing how it presents the target depending on its role, like a friend, mentor, or colleague [154]), or about whether it is intended to mimic behavior or how it should present the target in *new* situations the target has not itself been in [174]. Allowing an AI automaton whose target is a real individual to adapt to interactions or context can, however, exacerbate concerns such as about self-conception, identity fragmentation, or loss of agency [107, 154, 162].

Humanness: is the simulation intended to capture human-like characteristics? The mimicry or appearance of embodying human-like characteristics also influences how systems are perceived and interacted with, and the ethical concerns their deployment or use gives rise to [e.g., 64, 69, 71, 138, 213, 255]. This is the case even when there is no identifiable person or group being simulated, or when the simulation captures only general human-like attributes or behaviors [47, 52], as imbuing non-human agents with such qualities—e.g., appearance, intentions, motivations, goals—may end up objectifying and dehumanizing people [75, 127, 256], lead to anthropomorphic deception when users incorrectly believe they are talking to or interacting with a human rather than a machine [96, 205, 272], or lead users to develop material or emotional dependence on such agents [151, 166, 171] or a false sense of trust, safety, or familiarity [151, 182].

3.2 Intended Uses for the Simulation

Both the settings AI automatons are either developed for or are deployed and used in, as well as which and how various stakeholders are intended to benefit from interacting with these systems, determine *not only* their usefulness and how people perceive and interact with them, but also what risks their development and use might bring about [e.g., 110, 112, 154]. To help foreground possible design decisions that influence and are influenced by intended uses and goals, in our framework we consider four high-level considerations related to 1) whether the simulation is intended to *replace* the target, and 2) whether the system is set up in a way that enables others to *interact* with, 3) *observe*, or 4) *study* the simulation. These high-level considerations are primarily meant to make related design decisions more salient and are not mutually exclusive (e.g., a system could be designed both to replace the target as well as to allow the target to interact with their own simulation).

To replace: simulating in order to replace the target. Perhaps the most common concerns about AI automatons relate to how such systems could replace humans [e.g., 35, 38, 49, 112, 154, 210, 238], with growing beliefs that such systems could substitute humans in relationships [e.g., 6, 93, 289], and with some even declaring, for instance, that “the era of AI employees is here”—employees who “won’t complain about work-life balance” [218]. When and for what purposes the simulation’s target is replaced can thus color whether such replacement is seen as a *benefit* (e.g., when it enables the target to delegate unwanted or harmful tasks or to scale their work) or rather as a *concern* (e.g., the simulation of a target’s abilities is used to do their paid job and displace them).

To capture these differences, when one of the design goals is to replace the target, we distinguish between three different replacement goals for AI automatons: i) to *relieve*: intended to relieve the target (or others) from drudgery, possible harm, or activities that would be unethical or unsafe for the target (or others) to carry out. This is typically done by the operator to mitigate harm or provide relief for the benefit of the target (e.g., using simulated study participants to protect human subjects from harm [7, 135], or using AI news anchors to protect journalists from political retribution [209]); ii) to *substitute*: intended to be a stand-in or surrogate for the target when the target is unavailable, the target wants to delegate their tasks, or when the activity is impractical or impossible for the target to do, typically initiated by the target or with their knowledge, and to their benefit (e.g., responding to emails or messages on the target’s behalf [145, 160, 282] or standing in when human expert annotators are scarce [128]). For example, we consider the goal to be *substitution* when a target delegates a data annotation task to an AI automaton instead of doing it themselves, but *relief* when such annotations are done without the target’s involvement to protect them from possible harm from exposure to harmful content [e.g., 144],

as is often the case for automated assessments of hateful content [135]; and iii) to displace: intended to take over the place, position, or role traditionally occupied by the target to help an operator or interactor reduce costs, scale operations, increase speed, or enhance convenience, often to the detriment of the target (e.g., replacing human newscasters [186] or other human jobs [218] resulting in loss of opportunities or livelihood [35, 146]). This is typically done by an operator or interactor, often adversely impacting the target (or others) who may lack the means to mitigate the impacts (e.g., reduced wages, job loss, strained relationships). Here, much as with *relief*, the target is unlikely to have control over the simulation; but unlike *relief*, here the target is negatively impacted (e.g., by reduced wages, job loss [134]) and may not be able to mitigate such impacts.

To interact: the simulation is intended to be interacted with. AI automatons are increasingly developed for interaction [3, 11, 115, 166, 174, 200, 241], supporting a growing set of modes of interaction [122, 158, 171, 200]. This has also led to diverse conceptions of AI automatons and the roles they play, from collaborators to companions to coaches to judges (to name a few), which in turn influence how and for what purposes interactive AI automatons are developed, deployed, and used. The ability to interact with AI automatons in a growing number of settings has, however, also been accompanied by growing concerns, especially for those interacting with these systems, such as deskilling [116], emotional dependence [30, 151, 278], or addiction [278].

Interaction modes: the ways in which the simulation can be interacted with. When and how someone can interact with the simulation influences both the interaction dynamics as well as their perceptions of what is simulated and the consequences of doing so [68, 136, 156]. Operators' interactional goals are often guided by design considerations related to both the amount of freedom an interactor should have when engaging with a simulation (§3.3), as well as related to the types of actions the simulation is designed to carry out and for how long. The latter includes considerations about whether the simulation 1) supports open-ended or instead only more constrained, structured, or scripted interactions [99, 156, 174]; 2) allows only short-term versus longer-term interactions [78, 111]; 3) can be used in new situations rather than just reproducing past or known behaviors or situations [156, 174]; and 4) is intended to be generative and produce new behaviors or is rather intended to only be predictive or retrieval in nature [174]. While facilitating open-ended, long-term, generative interactions is more likely to lead to sensitive self-disclosures, emotional dependence, or psychosis [111, 119, 136, 176, 273], designing for more constrained, short-term interactions may also frustrate users for not recalling past interactions [62].

Stakes: the value interactors may derive from interacting with the simulation. When interaction is intended, a common leitmotif is wanting to support rather than replace humans [e.g., 61, 228, 271], with specific design choices motivated by varying aims for what the simulation is meant to do for an interactor [e.g., 53, 159, 166]. This is well-illustrated by the distinction drawn by Hofman et al. between cases where AI systems are intended to help users attain certain goals by serving as *steroids*—providing short-term performance boosts but risking deskilling in the longer term—*sneakers*—temporarily accelerating users' abilities—or *coaches*—helping improve users' own abilities rather than only helping them out in the moment. Such differences in how and what AI automatons are architected for can determine not only what impacts they may have on those interacting with them, but can also inform discussions about what trade-offs to strike between the value users may derive from these systems versus the adverse impacts these systems may have [e.g., 21, 125, 273].

To foreground differences in what AI automatons are developed for, we distinguish between several interaction goals: i) to enhance: improve or enhance the interactor's ability to complete a task or carry out an activity, without necessarily helping them also develop their skills (e.g., AI as steroids or as sneakers [112]); ii) to coach: train or teach the interactor to help them learn or improve their abilities and skills (e.g., act as a chess coach [174], practice with an automaton as an imagined audience [159, 166]); iii) to serve: provide specialized services to or for the interactor, which someone else would typically perform (e.g., a virtual nurse providing medical services to patients [44, 172]); iv) to connect: provide social or emotional support to interactors (e.g., companionship or friendship [32, 63]);

v) to entertain: entertain the interactor (e.g., gameplay [180], AI characters as a “new entertainment format” [55] or “designed to make you laugh, generate memes” [19]); vi) to accommodate: adapt or customize an AI automaton’s output or behavior to the target’s characteristics or needs, typically to increase familiarity or comfort, facilitate interactions, or provide a personalized experience (e.g., “Replika [...] learns [people’s] texting styles to mimic them” [120], “adapt the agent’s demeanor” [156], or customizing a generated voice depending on the interaction setting [37]); vii) to collaborate: act or serve as a collaborator for the interactor (e.g., machine or AI teammates [227, 286] or AI systems as “thought partners” [53]); and viii) to evaluate: assess the interactor, without necessarily being intended to help the interactor improve (e.g., a virtual interviewer developed to assess job applicants [143, 240]). These different settings are likely to exacerbate concerns in different ways; for instance, AI automatons deliberately designed to provide social and emotional support may be more likely to lead to emotional dependence, while those developed to entertain are more likely to be linked to concerns about addiction [e.g., 30, 252, 278].

Affinity: the intended or likely similarity between an interactor and a target. AI automatons designed for interaction can also vary in how much the target is intended to share some (or all) of the characteristics of those interacting with them. For instance, some simulations may only be designed to take on the interactor’s accent [91], while in other settings the target is the interactor [174, 260]. However, whether and how much a target shares the characteristics of an interactor or even those of the interactor’s *kith and kin* (e.g., such as having the same profession or demographic attributes)—either deliberately or accidentally—affects not only people’s perceptions of these systems but also how they interact with them [e.g., 122, 194]. Similarity is often desirable as it can facilitate familiarity and likability [90]; people tend to prefer and respond more positively to systems and representations they perceive as reciprocating or sharing some of their characteristics [37, 125, 126, 166, 193, 195], and even adjust their own behavior to virtual representations of themselves [249, 279]. Nevertheless, while in certain scenarios using someone’s accent or speaking style may help facilitate more high-quality interactions with a system, in others mimicking someone’s mannerisms or accent risks being perceived as mocking or stereotyping them [45, 89]. Virtual representations that come across as too eerie, creepy, or self-like have also been found to trigger adverse reactions [231], and people may want to customize representations to distance themselves from them or blur certain characteristics (e.g., gender or age [37]), or to project an idealized version of themselves or of others [e.g., 58, 286].

To showcase: the simulation is intended to be observed by others. AI automatons can also be designed to provide *non-interactive* spectator experiences, like watching or listening to AI-generated ads [114] or to image, video, or audio deepfakes [174]. Even when there are no direct interactions with the simulation, however, concerns can still arise depending on what is simulated, how it is simulated, and for what purposes, such as concerns about deception, misinformation, or reputational risks that may arise when the target is misrepresented [e.g., 28, 122, 154].

Stakes: the value spectators may derive from the simulation. Non-interactive AI automatons have also been developed for a variety of intended uses. For instance, in some non-interactive settings a target may be simulated to entertain an audience of spectators (e.g., AI-generated music [243], short films [220], or voices narrating stories [70, 122]). In other cases the AI automatons are meant to help train (e.g., a surgery demonstration [237, 259]) or to persuade those listening or watching (e.g., generated ads for marketing campaigns [39, 269]). As with interactive settings, such differences can sharpen risks in distinct ways: while copying one’s voice may prompt concerns about impersonation, consent, or appropriate compensation in most deployment settings [e.g., 122, 154], these concerns may be especially heightened when this is done for fraud or malicious persuasion [e.g. 125, 206].

Affinity: the intended or likely similarity between a spectator and a target. Analogous to the question of similarity between an interactor and a target (§3.2), the target can also vary in which and how many characteristics it shares with spectators, which can similarly color the spectators’ perceptions of and concerns about AI automatons. For instance, people may respond differently to a deepfake of themselves versus one of a public figure or a different everyday person, or versus a synthetic video of a fictitious individual or character [e.g., 33, 59, 73].

To study: simulating in order to study human or machine behavior or phenomena. Humans are also simulated for experimentation purposes to study theories about humans or the ability to simulate them. When the goal is to study either the targets or the automatons, we identify the following common goals of study: i) human behavior: when one or more targets and their interactions are simulated to understand populations and their possible behaviors; understand human beliefs, preferences, and values; or investigate any other human or social phenomena [e.g., 86, 132, 155, 214, 253, 262, 276, 281]; ii) the simulation: when one or more targets is simulated to test the ability to simulate the targets, or understand a simulation's properties (e.g., probe how well simulations of research participants align with human responses [15, 100, 224] or reproduce well-known experiments [8], or how well domain experts can be simulated [157]); iii) model or system performance: when one or more targets is simulated to anticipate how different stakeholders may interact with and use a model or system (e.g., simulate users interacting with a product to anticipate their needs [16, 223]).

3.3 Ownership & Control over the Simulation

Critical considerations in the deployment of AI automatons are also related to *by whom*, *when*, and *how* decisions are made about what is simulated. To help formalize how much *control* various stakeholders have over the scope and uses of the simulations, we adapt a conceptual framework for participation in AI that helps tease apart some of these considerations [66, 248],⁵ including 1) which stakeholders get to influence decisions, or *who is involved?* 2) what decisions these stakeholders get to influence, or *what is on the table?* and 3) in what ways they are able to influence decisions, or *what form does participation take?* The modes of participation Delgado et al. derived from existing literature further echo the different degrees of control or decision-making power different stakeholders could be given over what AI automatons are intended to do and how they can be used. Drawing on this work, we consider the following dimensions along which stakeholder control and related design considerations can vary:

Who is in control? Ethical stakes related to who and what about them is simulated, and how and by whom the simulation can be used, can feel different depending on *which stakeholders*—e.g., targets, interactors, spectators, operators—can participate in or influence decisions. For instance, an *operator* deciding who the simulation *target* is without their input or consent is more likely to raise concerns about issues with rights of publicity or consent circumvention about how their likeness or data is used [e.g., 13, 268]. Such concerns may be lessened when the *target* has full control over what about them is simulated and when their simulation can be used. These distinctions are critical as the ability to influence or make decisions about the development, deployment, or use of AI automatons also determines both 1) *with whom responsibilities lie* if and when these decisions lead to adverse impacts [122, 154, 156], and 2) the various stakeholders' ability to mitigate such impacts [e.g., 104, 221, 248, 270].

Type of control: what do they have control over? Different stakeholders may be able to influence or control different aspects of what is simulated, how and what the simulation is developed for, and even if it should be developed at all; such considerations about what stakeholders have control over—where the *locus of control* and responsibility lie—can help mitigate (or instead exacerbate) ethical concerns depending on how they limit or enable different stakeholders' influence over how AI automatons are architected and used. For instance, if the *target* only has control over what is simulated about them, but not over all the ways in which the simulation of their likeness, work, or abilities is used, they may still worry about reputational or discrimination risks and their ability to mitigate them. That is, a professional community or an actor may perhaps be comfortable with simulating their likeness to adjust a movie or documentary scene, but not for a video implying endorsement of a political candidate. People's preferences and concerns often depend on the context of use [42], and thus their ability to control when the reproduction of their likeness is being used. In addition, since developing a simulation

⁵While the scholarship and the body of work on participatory design is vast and growing, we primarily drew on the work and the framework by Delgado et al. on ownership and control, as that framework is based on a comprehensive survey of participatory design and AI.

is often data-intensive and requires a large corpus of digitized traces of a target’s behavior and likeness, questions about consent and control over one’s data are also particularly acute [184, 213].

To capture these distinctions, we consider if stakeholders have control over: i) whether a target is simulated: can influence or control who the *target* of the simulation is and if their simulation should be developed (e.g., users choose to build an AI version of themselves vs. a fictitious AI character [19]); ii) what about the target is simulated: can influence or control if the *target* as a whole is simulated or only some of their characteristics (e.g., users retain control over what is said on their behalf [217]); iii) how the simulation is developed: can influence or control how the simulation is implemented (e.g., how the target’s data can be used [113], mitigating only some consent-related concerns); iv) what the simulation is used for: can influence or control which deployment scenarios a simulation is developed for, how someone can interact with the simulation (if at all), or what tasks the simulation can perform (e.g., users can specify “topics to avoid” [19]). This can also include considerations about whether stakeholders can refuse interactions with a simulation; and v) who can interact with the simulation: can influence or control who has access to the simulation (e.g., only the target can interact with their simulation, or only adult users can interact with the simulation [113]).

Degree of control: how much control do they have? Stakeholders’ ability to influence the scope and use of simulations can also vary from no influence or control (e.g., fully autonomous agents that act without input), to being able to provide superficial feedback or input, all the way to having complete control—and thus able to make decisions about all aspects related to who and what is simulated, and when and how the simulations can be used. Stakeholders may also be able to influence or make decisions about the simulations only at certain points in an AI automaton’s development and deployment life-cycle. We thus consider two key dimensions of variation: 1) are stakeholders only able to provide feedback (*can influence*) or can they make decisions (*can control*)? and 2) when or where in the development and deployment life-cycle can stakeholders provide feedback or make decisions?

We operationalize these via five levels of stakeholder control: i) no control: no control or influence over the scope and use of the simulation—i.e., who and what about them is simulated, for what purpose, how the simulation can be interacted with, or how interactions with the simulation are used to adjust it (e.g., deepfakes of unsuspecting targets [17], or employees without control over being replaced by AI automatons [218]); ii) consulted: some influence over the scope and use of the simulation, typically by expressing discrete preferences or providing input at specific points in the development and deployment life-cycle (e.g., simulated chess coaches users can choose to use [174] but whose design they cannot influence); iii) included: can influence the scope and use of the simulation, typically through explicit feedback mechanisms implemented at most or all stages in the development and deployment life-cycle (e.g., indicate which type of messages an automaton can send on the target’s behalf [160]); iv) in control: can make some of the decisions about the scope and use of the simulations at specific points in the development and deployment life-cycle (e.g., choose gestures to animate in family photos [190] or specify who can interact with an automaton [e.g., 113, 187]); v) ownership: own the simulation or have full control over any part of the process used to create, deploy, or use the simulation, at any point in the development and deployment life-cycle (e.g., customize avatars for their own commercial purposes [9], or maintain “personal ownership and exclusive control over [their] digital image” [23]). A higher degree of control may help mitigate concerns about privacy or consent, but not about addiction or identity fragmentation.

3.4 Impacts from Simulating Humans

Recently, Meta took down AI accounts deemed creepy, inaccurate, and disrespectful [187], while Replika restored some of theirs after users expressed anguish from being separated from their AI partners [62, 170] due to an update disallowing certain uses. Indeed, concerns about how AI automatons may impact people and society govern both how people perceive and interact with them and the development of legal, ethical, and normative frameworks to guide and govern their use [e.g., 17, 174, 270, 284], which in turn influence what is built and deployed. Drawing

on existing literature on harm anticipation and taxonomization [31, 36, 122, 198], we foreground two key areas of consideration for design decisions: i) who may be affected by the development, deployment, and use of AI automatons (*impacted stakeholders*), and ii) how different stakeholders may be impacted (*adverse impacts*).

Impacted stakeholders: who is impacted? As with any AI system, reasoning about the implications of AI automatons requires careful consideration of all relevant stakeholders [36]. Automatons’ development, deployment, and use may impact not only *direct* stakeholders like those interacting with or the target of a simulation—e.g., family members believing their loved one was in an accident after interacting with a system imitating their voice [5], or a target’s identity being appropriated by third parties without consent [106, 216]—but also *indirect* stakeholders like individuals or communities associated with direct stakeholders even when they are not interactors (e.g., loved ones of a deceased target [185]), or even society at large (e.g., erosion of public trust [122]). Furthermore, even when given different interactors or operators with similar control over, e.g., an AI automaton designed to produce language in a minoritized variety, who the interactor or operator is might also impact concerns differently: when the operator is a speaker of the variety it may constitute *reclamation* or just ordinary use, whereas with a corporation or a non-speaker it may be seen as linguistic *appropriation*. Thus, as with design considerations related to *who is in control* (§3.3) of the development and deployment of AI automatons, differences in *which stakeholders*—e.g., targets, interactors, operators, or others—are involved and likely to be adversely impacted are influenced by *de facto* design decisions and *should* in turn influence those decisions.

Adverse impacts: how are they impacted? The risks to different stakeholders are similarly influenced by and *should* in turn influence how AI automatons are built and deployed [e.g., 42, 122, 154]. For instance, vulnerable individuals developing emotional attachment and trust towards an AI companion that results in them following harmful advice [24, 252] *should* perhaps minimally lead to these systems being designed to provide appropriate disclosures and reminders of interacting with an AI system to users, among other guardrails [30]. Similarly, concerns about misrepresentation *should* result in allowing a target to control what their simulations say and do in autonomous interactions [154]. Adverse impacts are also determined by how and when those risks are likely to arise or by possible *pathways to harm*—i.e., “causal chain[s] of events required for a harm to be realised” [54]. This includes considerations about how stakeholders get exposed to AI automatons (e.g., by being the target of, by interacting with, by operating, or by being denied access to an AI automaton [e.g., 122]), which system behaviors are more likely to give rise to certain adverse impacts [e.g., 42, 46], as well as the simulation’s role in heightening the risk of these impacts (e.g., by being the “perpetrator, instigator, facilitator, and enabler” of harms [284]).

4 Discussion and Concluding Remarks

AI automatons are developed and deployed in an ever-growing number of applications. The excitement around AI automatons—and their potential benefits—has, however, not been accompanied by a systematic understanding of the risks they pose. We developed our framework with the goal of helping developers and researchers recognize, make explicit, and analyze the design choices underpinning AI automatons. In so doing, we hope to support them in reflecting on the implications of those choices, including alternatives possibly available to them. While this work does not provide an ethical framework for deciding which choices are right, we believe that recognizing, explicitly articulating, and analyzing the design choices developers and researchers make is a prerequisite for establishing a basis for discussions about *how* to design and deploy AI automatons, including identifying redlines [e.g., 125] to guide decisions about *when* and *whether* to design and deploy AI automatons.

Being more explicit, reflexive, and intentional about design decisions. As we demonstrate, there is a wide range of design choices available to those seeking to develop AI automatons. Yet it is far from clear whether developers make such decisions by reflecting explicitly on the range of choices available to them—and then intentionally adopting those that best serve their goals. It is even less clear the degree to which developers reflect on how

different choices might affect the interests of those who serve as the target of the AI automaton, who interact with it, and even those who do not get to be a target of or interact with it. Our goal in developing this analytic framework was to foster greater recognition of the range of design choices available to developers such that they might make *better* choices. We did not set out to provide developers with a *how-to* guide for navigating the ethical issues that might arise in developing and deploying AI automatons. Given the many dimensions of possible variation—and the additional complexity that arises from their interaction—it is unlikely that there are general principles that can guide decision making across all possible configurations. But mapping out the vast space of design choices reveals that there are many paths that developers can and should consider—and that no one path is preordained. To support developers and researchers in explicitly articulating and documenting the choices they make when designing AI automatons, we provide in Appendix D a template for documenting those choices that mirrors our framework.

A foundation for more focused analyses of existing applications and for the design of more focused empirical studies. Our framework can also serve as the foundation for more targeted analyses of existing AI automatons, examining if seemingly similar applications actually vary along other dimensions—and if this variation seems to affect our ethical intuitions about their relative desirability. It could also help to reveal when there are consistent patterns in the configurations of certain applications, and such findings might invite further study looking into the possible reasons—e.g., technical, commercial, practical—why certain dimensions seem to vary consistently with each other, or if developers have clustered in a particular part of the space of choices. Similarly, the dimensions of variation identified in our analytic framework can inform the design of empirical studies that focus on interdependencies and interactions between different axes of design, and how those interactions might exacerbate or mitigate risks and concerns. Research subjects could be presented with different examples of AI automatons with carefully controlled design variations and interactions along specific dimensions (as illustrated in Appendix C), with the goal of assessing how their reactions or concerns differ when manipulating elements of the design configuration. While in this work we do not provide a clear link for how variations along certain design axes have a determinative impact on normative concerns, such future empirical studies are crucial both to better understand people’s reactions to the many possible ways of designing AI automatons and to provide evidence to better support researchers’ (including our own) ethical intuitions about different configurations’ desirability.

Challenges to articulating design decisions in practice. We argue that developers of AI automatons should clearly articulate their design choices to help stakeholders better understand the concerns that can arise from those choices, particularly those choices that may not be evident based on limited use of a system. At the same time, this is complicated in at least three ways. First, AI and machine learning research communities’ valorization of qualities such as generalizability [29] means that systems are regularly accompanied by broad claims of their capabilities or other characteristics—e.g., “general-purpose computational agents that replicate human behavior across domains” [201]—making it difficult for developers to precisely state, and other stakeholders to understand, who and which characteristics are simulated and for which purposes. Second, even if design decisions are well understood, it is often unclear what control current implementations of AI systems permit various stakeholders [e.g., 87, 280]. This is particularly salient when AI automatons are built atop foundation models “intended to be almost universally applicable” [248], raising questions about how to prevent an AI automaton from having knowledge and capabilities that the target might not have. Third, while the stakeholder categorization (§2.3) we use is based on the roles those stakeholders play in the design and use of AI automatons—with stakeholders being able to play multiple roles—the *operator* role may be inhabited by several different stakeholders among which there might be power differentials that can be hard to account for [e.g., 67, 181]. Individual developers or those designing or researching AI automatons might not have much decision power when it comes to the design and deployment of AI automatons. It thus remains an open question how and when illuminating design choices behind AI automatons can enable more effective intervention, governance, or resistance, a question future work should engage with.

Ethical Considerations, Adverse Impacts, and Positionality

Ethical considerations. In her commentary, Suchman argues that discussions of AI that hold it up as a self-evidently coherent, “stable and agential” entity elide important differences between various underlying techniques and between “speculative [...] projects and technologies in widespread operation,” uncritically reproducing beliefs in AI capabilities and making it difficult to carefully assess technologies and their impacts. While developing our framework, we thus *deliberately choose* not to focus on broad claims about AI automatons’ capabilities or speculations about what they can or cannot do, which can reflect perceptions or illusions of intelligence, agency, vitality, or other human-like qualities [69, 166, 241]; instead, we seek to address, as concretely and specifically as possible, the space of possible design goals and choices surrounding these AI automatons’ development, deployment, and use. While we introduce AI automatons as a broad category of objects, through our framework we aim to make it clear that they are not a singular or stable object, but that it can in fact be configured in many different ways, with equally as many impacts.

Adverse impacts. This choice, however, may also risk two adverse impacts: first, focusing on possible design goals and choices—i.e., what a developer wants or intends to build—may suggest that some designs are possible or even desirable to implement in practice, even when they may not be. In other words, this focus may overlook or even obscure questions and assumptions about why to even consider certain design goals or uses, why these goals and uses are desirable or defensible, what problems AI automatons are intended to address, and why developers believe AI automatons are a solution to those problems instead of other alternative ways—including ways that possibly involve entirely non-tech ways—to address the same problems. It might also obscure how in practice certain design choices might not be fully under the control of developers. Furthermore, foregrounding and speculating about a wider range of design choices might also risk drawing attention to possible system designs that might in fact heighten (rather than mitigate) existing concerns or even give rise to new ones—systems that perhaps should not be built. Second, not focusing on systems’ actual capabilities and implementations—i.e., what AI automatons can do and how they do it—also limits our ability to speak to what capabilities and implementations are already present in practice, and what their attendant impacts might be.

Positionality statement. AI automatons can be particularly evocative, sometimes even outright provocative. We were drawn to this topic, no doubt, by our own strong feelings about these emerging uses of AI systems. In attempting to identify the many possible design choices available to developers of AI automatons, we focused primarily on how different choices might create, exacerbate, or mitigate a broad range of harms. In doing so, we may have given the impression that we were motivated to work on this project because we have overwhelmingly or exclusively negative feelings about AI automatons. While we do have many reservations and concerns, that is not fully the case. We are open to the possibility that some types of AI automatons can serve beneficial purposes (e.g., human simulations for medical purposes), but we felt that the tech community’s excitement and enthusiasm around AI automatons and their applications has not been accompanied by a systematic understanding of their risks. While a rich literature has already developed that explores the many normative issues raised by different types of AI automatons, this work has remained somewhat disjointed and largely detached from the specific design choices available to developers of such systems. In other words, while we have seen encouraging and growing discussions about and policy efforts on the adverse impacts of AI automatons, many of these discussions and emerging policies continue to ignore what about these systems leads to those impacts. As a result, we suspect that it has been difficult for those developing such systems to understand both the full range of choices available to them and the implications of these choices. Our ultimate goal in writing this paper was not necessarily to arrive at a final judgment about the desirability of any given AI automaton, but to provide a framework and vocabulary to consider the merits of AI automatons in a more methodical, transparent, and inclusive manner that is appropriately informed by the range of risks these systems raise.

Generative AI Usage Statement

We did not use generative AI in the writing of this paper.

5 Author Contributions

AO proposed and conceptualized the framework and led the literature review and the writing of the manuscript. SLB, SB, and LE contributed to the conceptualization of the framework, to reviewing existing literature, and to the writing of the manuscript. All authors participated in discussions about the design considerations and their clustering (see Appendix B).

Acknowledgements

We are grateful to Manohar Swaminathan and the MSR FATE group for insightful discussions and feedback on older versions of this work. We are also grateful to Forough Poursabzi and the Psychological Influences of AI (Psi) project members for helping make the AI automats template clearer and more actionable.

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A AI Automatons Framework Overview

Table 1 provides a comprehensive overview of our analytical framework for the design of AI automatons, highlighting all design axes and considerations (§3.1–§3.4) along with examples from prior literature. It represents a more detailed breakdown of Figure 1.

B Methodological Approach Overview

To identify key design considerations for AI automatons we followed the following methodology:

Step 1: Identify and examine a purposive sample of related work. We collected an initial purposive sample of research papers that are concerned with the design, use, and impacts of AI automatons.

Step 2: Identify and (re)cluster design considerations. At each iteration, we inductively identified considerations involved in the design of AI automatons that already exist or had been considered in the literature, and examined and (re)clustered those considerations through collaborative discussion sessions with subsets of our research team.

Step 3: Identify additional literature. At each iteration, we also considered additional literature on the design, use, and impacts of AI automatons—via a mix of snowball and opportunity sampling—to help us identify further considerations involved in the design of deployed or researched AI automatons.

Step 4: Speculate about possible design variations. In our discussions, we also supplemented the examples of AI automatons mentioned in the literature with additional examples of both existing—particularly those featured in popular media—and speculative applications of AI automatons to help us deliberate about possible design variations and probe whether the framework was missing important axes of design variation.

Step 5: Expand the framework and repeat Steps 2 to 4 until no additional types of design considerations are identified. When we identified design considerations not already covered by our framework, we expanded and reorganized the framework. We stopped when the design considerations we identified in the previous steps did not result in changes to our framework.

Dimension	Description	Examples
Scope: What is the subject of the simulation?		
What is being simulated	Who is simulated?	specific individuals [174]; identity groups [48, 261]; subpopulation [15]; human-like AI or chatbots [180, 252]
– Target	– <u>Fictitious</u> : when the target or their characteristics are fictional, or not true or real	counterfeit people [25, 77]; AI character profiles [147]; AI generated users [55]; customized AI family [76]
	– <u>Real</u> : when the target or their characteristics represent an actual, real person or group	scam victims [41]; self [290]; real people [201]; famous people [250]
– Characteristics	What about the target is being simulated?	style transfer [50]; human-like typos [242]; childlike [106]
	– <u>Form</u> : the likeness, appearance, or style of the target	video game navigation [180]; entire personality [226]
	– <u>Content</u> : what the target might say or do	
How the target's characteristics are being simulated		
– Fidelity	How well or faithfully is the simulation intended to capture the target's characteristics?	algorithmic fidelity [15]; precision or fidelity of an AI model [174], clone [154], or avatar [40]
– Specificity	To what degree are the simulated characteristics unique to the target?	accounting for unique, individualized [...] characteristics [290]
– Completeness	To what degree is the simulation intended to capture the target fully or in its entirety?	conversational mannerisms [1]; digital doppleganger [165] or twin [202]
– Adaptability	To what degree is the simulation intended to evolve or adapt?	adaptability, versatility [183]; static clones [154]
– Humanness	To what degree is the simulation intended to capture human-like characteristics?	human-like sounds [232]; humanizing AI [251]; human emotions [34]
Intended Uses & Goals: How is the simulation intended to be used?		
To replace	Is the simulation intended to replace the target? For what purpose?	replace humans [51]; agents of replacement [95]
	– <u>Relieve</u> : relieve the target from drudgery or possible harm	automate drudgery [211]; overtake meaningless jobs [92]
	– <u>Substitute</u> : stand in for the target when the target wants to delegate a task	Dittos [156]; AI to talk to his wife for him [282]
	– <u>Displace</u> : take over the place, position, or role of the target to their detriment	AI employees [218]; replace survey respondents [7, 15], healthcare jobs [283]
To interact	Is the simulation intended to be interacted with?	interactive virtual humans [94] or deepfakes [115]
– Modes	How can the simulation be interacted with?	short vs. long-term [78]; open-ended vs. structured [99]
– Stakes	The value <i>interactors</i> may derive from interacting with the simulation	AI as steroids or sneakers [112]; AI-assisted writing [27, 124]
	– <u>Enhance</u> : improve or enhance the interactor's ability to carry out a task	AI as coach [112]; machine advisor [210]; simulated practice partners [163]; pedagogical agent [235]
	– <u>Coach</u> : train or teach the interactor to help help them learn or improve skills	AI nurse/therapist [44, 139, 172]; headshot generator [91]
	– <u>Serve</u> : provide a service to the interactor	companionable agents [234]
	– <u>Connect</u> : provide social or emotional support to interactors	"designed to make you laugh" [19]
	– <u>Entertain</u> : entertain the interactor	"adapt the agent's demeanor" [156]
	– <u>Accommodate</u> : adapt to the interactor's needs or characteristics	AI and machine teammates [227, 286, 291]; co-pilot [228]
	– <u>Collaborate</u> : act as a collaborator for the interactor	AI interviewer [143, 240]
	– <u>Evaluate</u> : assess the interactor, without the goal of helping them improve	affinity [40]; self-congruity [246]; self-dialogue [236]; self-clones [117]
– Affinity	The intended or likely similarity between an <i>interactor</i> and a target	
To showcase	Is the simulation intended to be observed by others?	AI-made ads [114]; synthetic media [106]
– Stakes	The value <i>spectators</i> may derive from the simulation	simulated voice narrating a story [70]
– Affinity	The intended or likely similarity between a <i>spectator</i> and a target	observe self-clones [290]
To study	Is the simulation for studying human or machine behavior or phenomena?	simulate social networks to study polarization [262]
	– Study human behavior	ability to simulate research communities [281]
	– Study the simulation	simulate users to test a product [16]
	– Study model/system performance	
Ownership & Control: Who makes decisions about the simulations?		
Who is in control? This can include targets, interactors, spectators, operators, or other third parties		user-generated [252]; developers [7]; stakeholders [154]
Type of control	What do they have control over?	user-controlled representations of self [104]
	– Over whether a target is simulated	control over simulating identifying attributes [221]
	– Over what about the target is being simulated	chatbot trained on one's own diary entries [118]
	– Over how the simulation is developed	control over downstream uses [270]
	– Over what the simulation is used for	control over to whom their AI can reply to [19]
	– Over who can interact with the simulation	
Degree of control	How much control do they have?	deepfakes of unsuspecting targets [17]
	– <u>No control</u> : no influence or control at any point in the development/deployment life-cycle	AI tutor personalized based on discrete feedback [203]
	– <u>Consulted</u> : some influence, but only at specific points in the development/deployment life-cycle	messages sent on the target's behalf [160]
	– <u>Included</u> : some influence at any point in the development/deployment life-cycle	animation of historical family photos [190]
	– <u>In control</u> : some control, but only at specific points in the development/deployment life-cycle	AI companion trained on developer's exes' profiles [142]
	– <u>Ownership</u> : full control at any point in the development and deployment life-cycle	
Impacts: What are the impacts of simulating humans?		
Stakeholders	Who is impacted?	parents of the interactor [24] or target [106];
Adverse impacts	How are they impacted?	physical harm [24]; perceived substitution risk [283]

Table 1. Overview of our conceptual framework. The examples either highlight prior work noting the design consideration, or illustrate variation along that dimension.

C Design Interactions and Impacts: Examples

We include two worked examples to illustrate how interactions between specific configurations or different design axes might heighten or mitigate concerns. Specifically, in the first example we reflect about 1) interactions between different simulation targets (who is being simulated?) and whether the simulation is intended to evolve or adapt over time (adaptability), while for the second example we examine 2) interactions between different simulation targets (who is being simulated?) and different types of control. For each of the two examples, we highlight some of the risks we envisioned that the specific interactions between different design axes might give rise to.

C.1 Interactions Between Variations in the Simulation Target and Variations in Adaptability

For the first example, we consider AI automatons designed for interaction, and specify the target in relation to the interactor (e.g., *self* means that the target is the same as the interactor, while *acquaintance* means the target is an acquaintance of the interactor). We are interested in how design choices about the target interact with design choices about whether the AI automaton is able to evolve or adapt.

Target	Interactions with Adaptability
Self	When the target is the same as the interactor, altering the characteristics of the target can exacerbate concerns about misrepresentations, or even concerns about their very own identity being exploited and displaced, eliciting negative emotional reactions.
Acquaintance	When the target is an acquaintance, changes in the simulated characteristics of the target are more likely to lead to confusion, disappointment, and erosion of the social relation between the interactor and the target if the target evolves in a way that is inconsistent to the beliefs that the interactor had about them. Fixing the identity of the target, or constraining the way the simulation of the target can evolve to reflect only changes to the actual target's experiences, might mitigate some of these concerns.
Public figure	When the target is a public figure, altering the characteristics of the target can heighten misrepresentation or reputational harms, if changes are inconsistent with the figure's public behavior or values. Depending on the figure's role (e.g., a government official), changes may increase the risk of misleading interactors and the public.
Unknown individual	When the target is an individual unknown to the interactor, concerns arising from altering the characteristics of the target may be reduced relative to other targets, as the interactor may not have existing beliefs about the target. Alterations, however, heighten risks related to the interactor developing misleading beliefs about the target and thus concerns about the target's reputation being damaged without appropriate notice. Unlike public figures, there might also be fewer protective mechanisms protecting them against the misuse or misrepresentation of their identities [117].
Fictitious character	When the target is a fictitious character, altering the characteristics of the target is less likely to result in concerns about identity fragmentation as it is not a real person, particularly if the target is a new fictitious character. Concerns about intellectual property, copyright, or misrepresentation might, however, still arise.
Group	When the target is a group, altering the characteristics of the target risks shifting the interactor's beliefs about that group of people. Risks of e.g., confusion, disappointment, and erosion of social relations are reduced (relative to other targets) so long as changes still result in AI automatons that interactors deem plausible. The need for consistency might also be lower, so long as the AI automaton seems to reflect at least some people in the group. However, as with self as target (especially if this is a group the interactor belongs to), changes in simulated characteristics could heighten concerns about misrepresentation, exploitation, or displacement.
Generic human	When the target is a generic human, or when simulating "a person" rather than "a specific person," one risk might be that interactors may take simulated characteristics (including changes) as true of people more generally.

C.2 Interactions Between Variations in the Simulation Target and Types of Control

For the second example, we consider the same configurations as for the example above (i.e., design for interaction, the target is specified in relation to the interactor), but for illustrative purposes we combine what the interactor has control over in two categories: 1) over who is simulated and how, and 2) over the development, deployment, or use of AI automatons.

Target	Interactions with Type of Control , such as control over ...	
	<i>who is simulated and what about them is simulated</i>	<i>the system development, deployment, or use</i>
Self	When the interactor is the target, many concerns for both the target and the interactor can be mitigated if the interactor has control over whether they are simulated and how. Depending on their choices, concerns about dehumanization, essentialization—reducing people to fixed, narrow representations, or instrumentalization—treating people as exchangeable or as a means to an end—remain.	When the interactor is the target, controlling development, deployment, or use might mitigate concerns about how others perceive, interact with, or use their simulation, but less so concerns about misrepresentation and risks of identity fragmentation, given the lack of control over whether they are simulated and how.
Acquaintance	Compared to target as self, risks are not fully mitigated for the target. Given the relationship between the target and the interactor, risks to the target might however be salient to the interactor. The interactor can still simulate the target without or even against their consent.	Controlling development, deployment, and use can reduce risks related to erosion of the social relation between the target and the interactor, but some concerns for the target such as related to consent remain.
Public figure, unknown individual, or group	Compared to target as self, risks are not fully mitigated for the target if the interactor has control but no incentives to prevent risks to them. The interactor can, for instance, choose to misrepresent the target or make choices that heighten concerns particularly salient for groups or public figures, such as stereotyping, erasure, or appropriation.	Compared to target as self, concerns remain for how others might interact with the target (e.g., in a way that normalizes certain behaviors towards the target or that stereotypes the target) or the setting the target might be simulated in.
Fictitious or generic human	Compared with prior configurations, without a specific or real target it might be unclear who else the simulation might affect.	While risks to the target might be less salient, in controlling development, deployment, or use the interactor may be able to make choices that reduce risks to themselves, such as over-reliance or emotional dependence.

D AI Automaton Documentation Template – [the name of the AI automaton]

Why is documenting design choices for AI automaton important? AI automaton have two properties that can heighten existing risks or introduce new ones: 1) they are intended or perceived to exhibit human-like characteristics, and 2) they reproduce or are intended to reproduce the characteristics of individuals, groups, or “generic” humans (i.e., when the characteristics are not intended to be and are not specific to a certain individual or group). Documenting and clarifying even implicit choices made when designing, building, and deploying AI automaton can help practitioners (e.g., developers, researchers, policy makers) better understand the implications (and possible risks) of those choices and lay grounds for exploring alternative design choices that could help mitigate risks. It can also help guide discussions about potential harms, alternative design choices, and possible mitigations. This template is meant to help you make design choices explicit in order to be able to reflect about those choices.

D.1 Description of [the name of the AI automaton]

Briefly describe your AI automaton. [add description here]

D.2 Scope: What is being simulated and how?

WHY you should document design choices related to the scope of what is simulated and how: Aspects related to who is simulated, what about them is simulated, or with how much accuracy they are simulated govern perceptions of and concerns about systems simulating people’s work, abilities, behavior, likenesses, or humanness. For instance, what is simulated and how the simulation is accomplished can influence perceptions of uncanniness and discomfort, particularly when aspects unique to an individual are simulated, when the simulation outputs appear eerily similar to what a human might do or look like, or when they capture one’s characteristics with high fidelity. Such properties of a simulation may also exacerbate other concerns like those about privacy violations and lack of appropriate consent, or may threaten someone’s sense of identity and agency.

WHAT about the simulation scope you should document: Fill in the right column with your responses. If some questions do not apply, please provide brief justifications.

Axis of design	Description for your AI automaton
What is being simulated?	
<i>1.a. Target: who is being simulated?</i>	
WHAT: Select and fill in the description for the option that applies. You can copy-paste the option and replace the text in the brackets. – Individuals: [name or describe the individual] – Groups: [name or describe the group; can be based on demographic, professional, or other group characteristics] – Generic human or human characteristics: [describe the characteristics being reproduced, e.g., a human voice] – A combination of multiple targets: [describe] Also describe whether some or all of the target characteristics are real, or whether some (or all) of them are fictitious. WHY & Examples: This is important as simulating fictitious characters that cannot reasonably be matched to a real person or group is less likely to raise concerns about, for instance, impersonation or lack of consent. If you are developing a general-purpose model or system that users can use to simulate a range of targets, please note this and reflect on the type of targets they could simulate using your AI automaton.	

<i>1.b. Characteristics: what about the target is simulated?</i> WHAT: List the characteristics being simulated. WHY & Examples: For instance, an application might be developed only to capture some of a target's characteristics and may also aim to do so only for specific actions or tasks a target may undertake. The simulation of different characteristics is likely to give rise to or heighten different sets of concerns.	
How are the target's characteristics being simulated?	
<i>1.c. Fidelity: how accurately is the simulation intended to capture these characteristics?</i> WHAT: Describe how well the application is intended to capture the target. WHY & Examples: The fidelity or accuracy with which an automaton reproduces a target's characteristics can impact its value or usefulness. On the one hand, high-fidelity simulations of an artist's style, work, or likeness are more likely to lead to copyright violations or infringement on their rights of publicity than low-fidelity ones. On the other hand, low-fidelity renderings can heighten risks related to misrepresentation, deception, and reputational harms.	
<i>1.d. Specificity: how identifiable or unique to the target are the simulated characteristics?</i> WHAT: Describe how unique the simulated characteristics are to the target. WHY & Examples: In addition to how faithfully a target or their characteristics are simulated, how unique these characteristics are to a target—e.g., in a way that uniquely represents or identifies them, or reproduces unique or rare abilities—is also critical to consider. Simulating unique characteristics is, for instance, more likely to limit the target's ability to maintain their individuality and capitalize on their own skills and talents.	
<i>1.e. Completeness: to what degree is the target intended to be captured fully or in its entirety by the simulation?</i> WHAT: Detail the extent to which the target intended to be reproduced—e.g., is it only some characteristics like their voice, or is the simulation also intended to reproduce someone visually or the way they make decisions? WHY & Examples: How many of a target's characteristics are simulated, or if it is intended to be simulated in its entirety, is another consideration that determines not only the deployment settings, but also how the automaton is perceived and interacted with, as well as how versatile the resulting automaton is in terms of the actions it can take. Simulations intended to be highly detailed and elaborate, exhaustive, or have high generality are likely to lead to more and heightened concerns. For instance, highly complex simulations of individuals are more likely to trigger concerns about objectification, dehumanization, displacement, or loss of individuality.	
<i>1.f. Adaptability: to what degree is the simulation able or intended to evolve or adapt?</i> WHAT: Describe whether and how the simulation is able or intended to evolve or adapt. WHY & Examples: While for some settings the simulations may be intended to remain static or reflect fixed snapshots of a target (e.g., cloning one's younger self to talk to them), other settings may require automatons to evolve 1) based on interactions, feedback, or new information (e.g., by learning from interactions), or 2) according to the target's own evolving self (e.g., to maintain accurate representations of the target). While a static snapshot or one that evolves separately from the target may misrepresent the target by presenting stale or inauthentic versions of them, settings when the simulation is intended to adapt to reproduce the target in new situations the target has not been in risks reproducing them in situations they would not have agreed to be in.	

1.g. Humanness: to what degree is the simulation intended to capture human-like characteristics?

WHAT: Describe whether a target is simulated with the goal of capturing general human-like characteristics, and what those characteristics are and why.

WHY & Examples: The mimicry or appearance of embodying human-like characteristics also influences how systems are perceived and interacted with, and the ethical concerns their deployment or use gives rise to, even when there is no identifiable person or group being simulated, or when the simulation captures only general human-like attributes or behaviors.

D.3 Intended uses: For what purposes is the target simulated?

WHY you should document design choices related to how the AI automaton is used: The settings AI automaton are developed for and deployed in, how AI automaton are used, and which and how various stakeholders may benefit from interacting with these systems determine not only their usefulness and how people perceive and interact with them, but also what risks their use might bring about. To help foreground possible design choices that influence and are influenced by intended uses and goals, we ask you to reflect on design choices related to whether the simulation is intended to replace the target, and whether the system is set up in a way that enables others to interact with, observe, or study the simulation.

WHAT about the use of AI automaton you should document: Fill in the right column with your responses. If some questions do not apply, note N/A or provide brief justifications as appropriate.

Axis of design	Description for your AI automaton
<i>2.a. To replace: is the simulation intended to replace the target?</i> WHAT: Describe whether the target is intended to be replaced and, if so, for what purposes. WHY & Examples: When and for what purposes the simulation's target is replaced can color whether such replacement is seen as a benefit (e.g., when it enables the target to delegate unwanted tasks or scale their work) or rather as a concern or risk (e.g., the simulation of a target's abilities is used to do their paid job and replace them). As you reflect on this, try to distinguish between these different types of replacement goals: • to relieve the target from drudgery or possible harm • to substitute by acting as a stand-in or surrogate for the target when the target is unavailable or wants to delegate a task • to displace by taking over the place, position, or role traditionally occupied by the target to help reduce costs, scale operations, increase speed, or enhance convenience, possibly to the detriment of the target	
<i>2.b. To interact: is the simulation intended to be interacted with?</i> – <i>Interaction modes: in what ways the simulation can be interacted with?</i> WHAT: Describe how the simulation can be interacted with. WHY & Examples: When and how someone can interact with the simulation influences both the interaction dynamics as well as their perceptions of what is simulated and the consequences of doing so. Operators' interactional goals are guided by design considerations related to both the amount of freedom an interactor should have when engaging with a simulation, as well as about the types of actions the simulation is designed to carry out and for how long.	

– *Stakes: what value are the interactors intended to derive from interacting with the simulation?*

WHAT: Describe what the simulation is intended to do for users.

WHY & Examples: Differences in what AI automatons are architected for can determine not only what impacts they may have on those interacting with them, but can also inform discussions about what trade-offs to strike between the value users may derive from these systems versus the adverse impacts these systems may have. Examples of goals or intended uses can include enhancement, coaching users, providing social and emotional support to users, entertaining users, collaborating with users, and evaluating users.

– *Affinity: what is the intended or likely similarity between the interactor and a target?*

WHAT: Describe how similar to the interactor the target is intended to be.

WHY & Examples: Whether and how much a target shares the characteristics of an interactor or even those of the interactor's kith and kin (e.g., such as having the same profession or demographic characteristics)—either deliberately or accidentally—affects not only people's perceptions of these systems but also how they interact with them.

2.c. To showcase: is the simulation intended to be observed by others?

– *Stakes: what values are the spectators or observers intended to derive from the simulation?*

WHAT: Describe what the simulation is intended to do for those who can observe it.

WHY & Examples: Simulations can also be intended to provide non-interactive experiences, like watching or listening to AI-generated ads or a video/audio deepfake. Even when there are no direct interactions with the simulation, concerns can still arise depending on what is simulated, how it is simulated, and for what purposes.

– *Affinity: what is the intended or likely similarity between the spectator and a target?*

WHAT: Describe how similar to the spectator the target is intended to be.

WHY & Examples: Whether and how much a target shares the characteristics of an spectator or even those of the spectator's kith and kin (e.g., such as having the same profession or demographic characteristics)—either deliberately or accidentally—affects not only people's perceptions of these systems but also how they interact with them.

2.d. To study: is the simulation intended for studying human or machine behavior or phenomena?

WHAT: Describe what the simulation is intended to help study.

WHY & Examples: Humans are also simulated for experimentation purposes in order to study theories about humans or about the ability to simulate them. When the goal is to study either the simulation targets or the simulations themselves, common goals typically include one of the following (or a mix):

- study human behavior
- study the simulation
- assess model/system performance

D.4 Other design, development, and deployment considerations

WHY you should document any other design choices: Critical considerations in the deployment of AI automatons are also related to by whom, when, and how decisions are made about what is simulated, how the simulation can be interacted with, who can interact with the simulation and when, and the process of developing and deploying the simulations. These decisions often relate to how much control or ownership various stakeholders have over the scope and uses of the simulations or over resulting artifacts (models, data, generated outputs, or produced artifacts). They also relate to notions of consent and whether there are any mechanisms for recourse and redress, as well as aspects related to the data and methods being used.

WHAT about other choices you could document: Fill in the right column with your responses. If some questions do not apply, please provide brief justifications.

Axis of design	Description for your AI automaton
<i>3.a. Control and ownership: who decides what can be simulated and how the simulation is used?</i> WHAT: Describe how it is decided what is simulated and how the simulation is used. WHY & Examples: Aspects related to <i>by whom</i>, <i>when</i>, and <i>how</i> decisions are made about what is simulated can both heighten or mitigate concerns. They also relate to notions of consent and whether there are any mechanisms for recourse and redress. The sections below will help you further expand on three dimensions of control-related considerations.	
– <i>Stakeholders: Who is in control?</i> WHAT: Describe which stakeholders have control over the simulation and its deployment and use. If none, please briefly explain. WHY & Examples: Concerns about who and what about them is simulated, and how and by whom the simulation can be used, can be mitigated or heightened depending on which stakeholders—e.g., targets, users, operators—can participate in or influence decisions. Thus, stakeholders typically include one or a mix of the following: • The targets or those whose likeness, characteristics, abilities, work, behavior or humanness are simulated • The users or those able to observe or interact with the simulation • The operators or those responsible for developing/deploying the system • Other stakeholders (e.g., family of a target, the creator of a target for fictional targets)	
– <i>Type of control: What do these stakeholders have control over?</i> WHAT: Describe briefly the type of control different stakeholders have. WHY & Examples: Different stakeholders may be able to influence or control different aspects of what is simulated, how and what the simulation is developed for, and even if it should be developed at all; choices about what stakeholders have control over—or where the locus of control and responsibility lie—can help mitigate (or instead exacerbate) concerns depending on how they limit or enable different stakeholders' influence over how simulations are architected and used. Thus, such aspects typically include one or a mix of the following: • Whether a target is simulated • What about the target is simulated • How the simulation is developed • What the simulation is used for (this can include both aspects related to intended uses, as well as modes of interaction) • Who can interact with the simulation	

– <i>Degree of control: How much control do different stakeholders have?</i> WHAT: Describe how much control each type of stakeholder has over any of the design, development, and deployment choices mentioned above. Describe any dedicated processes or mechanisms to provide control over what is being simulated and how. WHY & Examples: Stakeholders' ability to influence the scope and use of AI automatons can vary from no influence or control (e.g., fully autonomous agents that act without input), to being able to provide superficial feedback or input, all the way to having full control—and thus able to make decisions about any aspects related to the design, development, deployment and use. Stakeholders may also be able to influence or make decisions only at certain points in the development/deployment life-cycle. Consider these different levels of control: • No control: no control or influence over the scope and use of the simulation • Consulted: some influence over the scope and use of the simulation, e.g., by expressing discrete preferences or providing input at specific points in the development/deployment life-cycle • Included: can influence the scope and use of the simulation, e.g., by explicit feedback mechanisms at most/all stages in the development/deployment life-cycle • In control: can make some of the decisions about the scope and use of the simulation at specific points in the development/deployment life-cycle • Ownership: own the simulation or have full control over any part of the process used to create, deploy, and use the simulation	
<i>3.b. Other development and deployment considerations: what other choices related to the development and deployment of AI automatons might impact interactions and perceptions?</i> WHAT: Describe any other development and deployment considerations that you believe might impact how AI automatons are perceived and interacted with, as well as any potential adverse outcomes. WHY & Examples: Such considerations can include choices related to the methods and the data being used to develop the AI automaton.	

D.5 Impacts: What are the impacts of simulating humans?

WHY you should document possible adverse impacts: Concerns about how AI automatons may impact people and society govern both how people perceive and interact with them, as well as the development of legal, ethical, and normative frameworks to guide and govern their use, which in turn influence or should influence what is built and deployed. Key considerations here include who is being affected and how they are affected.

WHAT about adverse impacts you should document: Fill in the right column with your responses. If some questions do not apply, please provide brief justifications.

Axis of design	Description for your AI automaton
<i>4.a. Stakeholders: Who is impacted?</i> WHAT: Describe who is being impacted by the design, development, deployment, or use of the AI automaton. WHY & Examples: Reasoning about the design and implications of the AI automaton requires careful consideration of all relevant stakeholders. Automatons' development, deployment, and use may impact not only <i>direct</i> stakeholders like those interacting with or the target of a simulation—e.g., family members believing their loved one was in an accident after interacting with a system imitating their voice, or a target's identity being appropriated by third parties without consent—but also <i>indirect</i> stakeholders like individuals or communities associated with direct stakeholders even when not interactors (e.g., loved ones of a deceased target), or even society at large (e.g., erosion of public trust).	

4.b. Adverse impacts: How are different stakeholders impacted? WHAT: Describe how each stakeholder you identified may be impacted by the design, development, deployment, or use of the AI automaton. WHY & Examples: Risks to different stakeholders are influenced by and should in turn influence how simulations are built and deployed. For instance, vulnerable individuals developing emotional attachment and trust towards an AI companion that results in them following harmful advice should perhaps minimally lead to these systems being designed to provide appropriate disclosures and reminders of interacting with an AI system to users, among other guardrails. Similarly, concerns about misrepresentation should result in allowing a target to control what their simulations say and do in autonomous interactions. Adverse impacts are also determined by how and when those risks are likely to arise or by possible <i>pathways to harm</i>. This includes considerations about how stakeholders get exposed to AI automaton (e.g., by being the target of, by interacting with, by operating, or by being denied access to an AI automaton), which system behaviors are more likely to give rise to certain adverse impacts, as well as the simulation’s role in heightening the risk of these impacts),	
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